

Restaurant Revenue & Demand Analytics

IBM Project Report

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1. Executive Summary

This project delivers a complete, end-to-end restaurant revenue and demand analytics solution built for the IBM SkillsBuild Final Project submission. The objective is to transform a year of raw restaurant sales transaction records into actionable business intelligence — covering data quality, feature engineering, KPI dashboards, exploratory analysis, and machine learning demand prediction.

Dataset: 10,000 transaction records from 50 restaurants across 14 menu items, spanning 1 January 2024 to 1 January 2025. The data covers five restaurant types (Food Stall, Cafe, Kopitiam, Casual Dining, Fine Dining), three meal types, and three weather conditions.

Analytical Approach: The pipeline applies data cleaning, financial feature engineering (revenue, gross margin, margin percentage), and thorough exploratory data analysis across dimensions including menu items, restaurant types, meal periods, time trends, promotions, special events, and weather.

Machine Learning: A Random Forest Regressor was trained to predict `quantity_sold` using a time-aware train/test split. The model achieved an R^2 of 0.758, MAE of 78.53 units, and RMSE of 109.43 units — explaining approximately 75.8% of demand variance.

Dashboard: A four-tab interactive Streamlit dashboard presents all KPIs, trends, EDA charts, ML results, and dynamically generated business insights — all updating in real time based on user-selected filters.

2. Problem Statement

Restaurants operate in a highly competitive, margin-sensitive environment where understanding revenue drivers and accurately forecasting demand can be the difference between profitability and loss. Despite generating large volumes of transaction data, many restaurant operators lack the analytical tools to answer basic questions such as:

- Which menu items generate the most revenue and the highest gross margins?
- How do external factors — promotions, special events, weather — correlate with demand?
- Can historical data be used to predict future demand and support inventory planning?
- Which restaurant types and meal periods are most profitable?
- Are there pricing inconsistencies that should be investigated?

This project addresses these questions by building a data pipeline that loads, cleans, and analyses real restaurant transaction data, then produces an interactive dashboard and a predictive model — all in a single deployable Python file.

3. Dataset Description

3.1 Overview

The dataset file `restaurant_sales_data.csv` contains 10,000 rows and 13 columns. Each row represents one menu-item transaction record at a specific restaurant on a specific date and meal period.

Property	Value	Notes
Total Rows	10,000	One record per menu item per restaurant per date-meal combination
Total Columns	13	Mix of dates, identifiers, prices, quantities, and flags
Date Range	1 Jan 2024 – 1 Jan 2025	Full calendar year (366 days; 2024 is a leap year)
Unique Restaurants	50	IDs 1–50; each maps to exactly one restaurant type
Unique Menu Items	14	Consistent spellings throughout; no name variants

3.2 Column Descriptions

Column	Data Type	Description
date	Date (MM/DD/YYYY)	Transaction date; parsed to datetime64 during cleaning
restaurant_id	Integer (1–50)	Unique identifier for each of the 50 restaurants
restaurant_type	Categorical	One of: Food Stall, Cafe, Kopitiam, Casual Dining, Fine Dining
menu_item_name	Categorical	One of 14 menu items (e.g. Kaya Toast Set, Chicken Chop, Laksa)
meal_type	Categorical	Breakfast, Lunch, or Dinner
key_ingredients_tags	Free text	Comma-separated ingredient list (not used in ML features)
typical_ingredient_cost	Decimal (RM)	Estimated raw ingredient cost per unit
observed_market_price	Decimal (RM)	Reference/competitor market price for the same item
actual_selling_price	Decimal (RM)	Price at which the restaurant sold the item
quantity_sold	Integer (≥ 0)	Units sold on that date/meal/restaurant combination
has_promotion	Boolean (TRUE/FALSE)	Whether a promotional offer was active
special_event	Boolean (TRUE/FALSE)	Whether a special event occurred on that date
weather_condition	Categorical	One of: Sunny, Cloudy, Rainy

4. Data Quality Audit & Cleaning

4.1 Quality Audit Findings

Check	Result	Action Taken
Missing values	0 across all 13 columns	No imputation required
Duplicate rows (exact)	0	No rows dropped
Duplicate transaction keys*	366 groups (755 rows)	Retained as instructed; flagged in audit report
Zero quantity sold	502 rows (5.0%)	Retained; excluded from average-demand calculations
Negative prices	0	No action required
Below-cost selling	0 (selling price always $\geq$ ingredient cost)	No action required
Categorical inconsistencies	None (spellings fully consistent)	No standardisation needed
Invalid dates	0	All dates parse successfully

* Duplicate transaction keys: same date + restaurant_id + menu_item_name + meal_type combination. These may represent multiple sittings. Retained without deduplication.

4.2 Cleaning Steps Applied

The following cleaning operations were applied in `clean_data()`:

- Date parsing: date column converted from M/D/YYYY text to Python `datetime64` using `pd.to_datetime()`.
- Boolean conversion: `has_promotion` and `special_event` converted from text strings 'TRUE'/'FALSE' to native Python booleans to prevent silent comparison failures.
- Numeric casting: `typical_ingredient_cost`, `observed_market_price`, and `actual_selling_price` cast to `float64`; `quantity_sold` and `restaurant_id` cast to `int64`.
- Whitespace stripping: All categorical text columns trimmed.
- Temporal sorting: Dataset sorted by date ascending before the time-aware train/test split for ML.

5. Feature Engineering

The `engineer_features()` function enriches the cleaned dataset with the following derived columns, which are used in all subsequent KPI calculations, EDA, and machine learning.

5.1 Financial Features

Feature	Formula	Purpose
revenue	$\text{actual_selling_price} \times \text{quantity_sold}$	Total money earned per transaction row
estimated_cost	$\text{typical_ingredient_cost} \times \text{quantity_sold}$	Estimated raw ingredient cost for the units sold
estimated_gross_margin	$\text{revenue} - \text{estimated_cost}$	Estimated profit before overheads

Feature	Formula	Purpose
margin_percentage	$\text{estimated_gross_margin} / \text{revenue} \times 100$	Gross margin as a percentage; NaN when revenue = 0
markup_ratio	$\text{actual_selling_price} / \text{typical_ingredient_cost}$	Price markup over ingredient cost

5.2 Pricing Features

Feature	Formula	Purpose
price_vs_market	$\text{actual_selling_price} - \text{observed_market_price}$	Positive = restaurant charges above market; negative = below

5.3 Date Features

Feature	Source	Purpose
year	date.dt.year	Calendar year (all 2024 in this dataset)
month	date.dt.month	Month number (1–12); used as ML feature
month_name	month mapped to Jan–Dec	Human-readable label for charts
day	date.dt.day	Day of the month
day_of_week	date.dt.dayofweek (0=Mon ... 6=Sun)	Used in EDA and ML features
week	date.dt.isocalendar().week	ISO week number; used as ML feature
is_weekend	day_of_week in {5, 6}	Boolean flag; significantly correlates with higher demand

Note: No hour feature was created because the dataset contains no time-of-day information — only meal_type (Breakfast/Lunch/Dinner) as a period indicator.

6. KPI Analysis

The following top-level KPIs were calculated from the full cleaned dataset (all 10,000 rows, full year).

KPI	Value	Notes
Total Revenue	RM 32,713,418	Sum of actual_selling_price × quantity_sold across all rows
Total Quantity Sold	2,860,004 units	Total items sold across all restaurants, items, and dates
Estimated Gross Margin	RM 24,884,683	Total revenue minus total estimated ingredient cost
Average Selling Price	RM 18.86	Mean actual_selling_price per transaction row
Average Gross Margin %	74.2%	Average across all non-zero revenue rows
Number of Restaurants	50	IDs 1–50
Number of Menu Items	14	Distinct menu_item_name values
Transaction Records	10,000	Total rows; includes 502 zero-quantity rows

The overall average gross margin of 74.2% is high, which is typical of food service businesses where ingredient costs are a small fraction of the selling price. This estimate covers ingredient costs only and does not account for labour, rent, utilities, or other overheads.

7. Exploratory Data Analysis

7.1 Revenue by Menu Item

Menu Item	Total Revenue (RM)	Units Sold	Avg Selling Price (RM)
Kaya Toast Set	5,240,720	544,101	10.07
Chicken Chop	4,762,333	109,641	47.26
Spaghetti Carbonara	4,511,648	93,114	53.48
Beef Rendang	2,919,473	66,038	48.14
Nasi Lemak	2,423,090	226,795	11.22
Cendol	2,213,847	304,622	7.86
Tandoori Chicken	2,139,956	91,048	25.53
Teh Tarik	1,954,970	643,076	3.16
Iced Lemon Tea	1,400,942	235,325	6.28
Mushroom Soup	1,392,305	70,164	21.46
Char Kway Teow	1,264,687	64,279	21.40
Chicken Rice	1,171,895	77,949	16.09
Roti Canai	662,169	286,505	2.43
Laksa	655,384	47,347	15.20

Key observation: Kaya Toast Set leads revenue by volume (highest unit count at 544,101 units), while Spaghetti Carbonara leads on average price (RM 53.48). High-volume low-price items (Teh Tarik, Roti Canai) may still be valuable as traffic drivers even if their individual revenue contribution is smaller.

7.2 Revenue by Restaurant Type

Restaurant Type	Total Revenue (RM)	Revenue Share
Food Stall	9,686,508	29.6%
Fine Dining	9,061,345	27.7%
Casual Dining	8,900,088	27.2%
Cafe	4,470,601	13.7%
Kopitiam	594,875	1.8%

Food Stalls lead in total revenue despite having lower prices, driven by very high unit volumes. Fine Dining achieves similar total revenue with far fewer transactions at much higher average prices. Kopitiam has the smallest share at 1.8%.

7.3 Gross Margin by Menu Item

Menu Item	Avg Gross Margin %
Mushroom Soup	82.4%
Chicken Chop	81.6%
Spaghetti Carbonara	81.5%
Iced Lemon Tea	79.6%
Beef Rendang	78.5%

Menu Item	Avg Gross Margin %
Nasi Lemak	75.0%
Char Kway Teow	74.6%
Chicken Rice	73.1%
Cendol	72.6%
Tandoori Chicken	71.2%
Kaya Toast Set	70.5%
Teh Tarik	69.8%
Laksa	68.9%
Roti Canai	66.2%

All menu items maintain a gross margin above 66%. Mushroom Soup leads at 82.4%. These figures represent ingredient-cost margins only and do not include labour, rent, or other overhead costs.

7.4 Meal Type Analysis

Meal Type	Avg Demand (units/row)	Total Units Sold
Breakfast	460.3	1,157,602
Lunch	247.6	863,103
Dinner	209.9	839,299

Breakfast records the highest average demand per row (460.3 units) — nearly double the dinner average (209.9). This is driven by high-volume, low-price breakfast items such as Kaya Toast Set and Teh Tarik at Food Stalls and Kopitiams.

7.5 Monthly Revenue Trend

Month	Total Revenue (RM)
January 2024	2,977,609
February	2,773,600
March	2,715,702
April	2,720,189
May	2,740,256
June	2,579,611
July	2,563,408
August	2,923,408
September	2,382,221
October	2,675,806
November	2,695,840
December	2,965,767

Revenue is broadly stable across all 12 months (range RM 2.38M–RM 2.98M). There is no sharp seasonal peak. January, August, and December are marginally stronger; September is the lowest month. The project uses this monthly stability to support robust ML generalisation across time.

7.6 Weekday vs Weekend

Weekend records average 370.0 units per row versus 253.1 on weekdays — a 46.2% higher demand on weekends. This is one of the strongest signals in the dataset and is ranked 7th by the Random Forest model in feature importance.

7.7 Promotions

Condition	Avg Demand (units/row)	Row Count
No Promotion	251.4	8,502
Promotion Active	482.2	1,498

Rows with an active promotion show 91.8% higher average demand (482.2 vs 251.4 units/row). This is the strongest observed demand correlate in the dataset. Caution: this is an observational association — promotions may be selectively applied on already high-demand days, which would inflate the measured effect.

7.8 Weather Effect

Weather Condition	Avg Demand (units/row)
Sunny	298.0
Cloudy	281.4
Rainy	263.3

Sunny weather correlates with the highest average demand (298.0 units/row). Rainy weather shows the lowest (263.3) — a 13.2% difference between the extremes. Weather is a weaker signal than promotions or meal type, but is included in the ML feature set.

8. Demand Analysis

The following demand drivers were identified through EDA, ranked by the size of the observed effect on average quantity_sold per row:

- Promotions — 91.8% higher demand when has_promotion is active (482.2 vs 251.4 units/row). Strongest observed correlate. Present in 15.0% of rows.
- Weekends — 46.2% higher demand on Saturdays and Sundays (370.0 vs 253.1 units/row).
- Breakfast meal type — 460.3 avg units/row vs 209.9 for Dinner. High-volume, low-price breakfast staples drive overall totals.
- Special events — 28.8% higher demand on special event days (363.0 vs 281.9 units/row). Present in 5.1% of rows.
- Weather — Sunny days average 298.0 vs 263.3 on Rainy days (13.2% difference). A weaker but real signal.

Note: All figures above are observational correlations from historical data. They do not establish causation, and confounding factors (e.g., promotions being more likely on weekends) may influence the observed associations.

9. Machine Learning

9.1 Objective

The machine learning component predicts quantity_sold (demand) for a given restaurant, menu item, meal type, date, pricing, and contextual conditions. The target variable is quantity_sold — the integer number of units sold per transaction row.

9.2 Algorithm

Algorithm: RandomForestRegressor (scikit-learn)

Configuration: n_estimators=150, max_depth=12, min_samples_leaf=5, random_state=42, n_jobs=-1

- Random Forest was chosen for several reasons:
- Handles mixed numeric and encoded categorical features natively
- Robust to outliers (the max quantity_sold is 1,668)
- Provides feature importance scores for interpretability
- Does not require feature scaling
- Produces stable results across random seeds

9.3 Train/Test Split

A time-aware split is used instead of random shuffling. The dataset is sorted by date and split at 80%: the first 8,000 rows (up to approximately October 2024) form the training set, and the remaining 2,000 rows (approximately November 2024 – January 2025) form the test set. This simulates a realistic production scenario where the model is trained on past data and evaluated on unseen future data.

Split	Row Count
Training set (first 80% by date)	8,000 rows
Test set (last 20% by date)	2,000 rows

9.4 Feature Set & Leakage Prevention

The following 14 features are used. Revenue, estimated_cost, estimated_gross_margin, and margin_percentage are deliberately excluded to prevent target leakage — all four are derived from quantity_sold itself and would give the model access to the answer during training.

Feature	Type	Feature Importance (RF)
actual_selling_price	Numeric	0.5662 (1st)
typical_ingredient_cost	Numeric	0.0768 (2nd)
menu_item_name (encoded)	Encoded categorical	0.0762 (3rd)
meal_type (encoded)	Encoded categorical	0.0655 (4th)
observed_market_price	Numeric	0.0623 (5th)
day_of_week	Integer (0–6)	0.0411 (6th)
is_weekend	Binary flag	0.0331 (7th)
restaurant_type (encoded)	Encoded categorical	0.0287 (8th)
restaurant_id	Integer (1–50)	0.0159 (9th)
week	Integer (1–53)	0.0156 (10th)
month	Integer (1–12)	(lower importance)
weather_condition (encoded)	Encoded categorical	(lower importance)
has_promotion	Binary (0/1)	(lower importance)
special_event	Binary (0/1)	(lower importance)

The dominant feature (actual_selling_price, importance = 0.5662) reflects the fact that price strongly differentiates items. Higher-priced Fine Dining items sell in small volumes while low-price Food Stall items sell in very large volumes. This is a natural feature of the data structure, not a causality claim.

10. Model Evaluation

The model was evaluated on the held-out test set (2,000 rows covering the last 20% of dates). All metrics below are computed from actual predictions on unseen data — no data leakage.

Metric	Value	Interpretation
MAE (Mean Absolute Error)	78.53 units	On average, predictions are off by 78.5 units from the true quantity sold
RMSE (Root Mean Square Error)	109.43 units	Higher than MAE due to sensitivity to large errors; large outliers exist in the dataset
R ² (Coefficient of Determination)	0.758	The model explains 75.8% of the variance in quantity_sold on the test set

Interpretation: An R² of 0.758 is a strong result for a demand prediction task where quantity_sold ranges from 0 to 1,668 units. The MAE of 78.5 units represents an average error of approximately 27% of the mean quantity sold across the test set. The gap between MAE and RMSE indicates the presence of some large individual errors, consistent with the high-volume outlier rows (e.g., promotions or special events with >1,000 units).

The Actual vs Predicted Demand chart (available in the Streamlit dashboard, Demand & Operations tab) shows each test-set transaction as a scatter point. Points close to the red dashed 'perfect prediction' line represent accurate predictions. Most predictions are clustered tightly around the line, with some scatter at very high demand values — consistent with the RMSE/MAE gap.

11. Business Insights

All insights below are derived directly from the actual dataset. They describe observed patterns and correlations in historical data. None of these findings establish causation.

- Kaya Toast Set is the top revenue item at RM 5,240,720, driven by the highest unit volume (544,101 units) at a low average price of RM 10.07. Its strong performance is driven by volume, not premium pricing.
- Spaghetti Carbonara and Chicken Chop are the two highest-priced items (avg RM 53.48 and RM 47.26 respectively), yet both appear in the top 3 by total revenue. These items are profitable at scale even with lower unit volumes.
- Mushroom Soup has the highest average gross margin at 82.4%, despite being only the 10th item by total revenue. It is the most margin-efficient item per unit sold.
- Food Stalls generate the largest revenue share (29.6% / RM 9.69M) through extremely high volumes at low prices. Fine Dining achieves nearly the same share (27.7% / RM 9.06M) through a much smaller number of high-value transactions.
- Promotions are associated with 91.8% higher average demand (482.2 vs 251.4 units/row). This is the largest observed effect size in the data. The relationship is

correlational — it cannot be stated that promotions cause higher demand without a controlled experiment.

- Weekend demand is 46.2% higher than weekday demand (370.0 vs 253.1 units/row). This holds consistently and is the second-most important factor observed.
- Breakfast is the highest-demand meal type (460.3 avg units/row), largely due to high-volume Food Stall items served in the morning. Dinner is the lowest (209.9).
- September is the lowest revenue month (RM 2,382,221) and July the second-lowest (RM 2,563,408). January, August, and December are the strongest months. No extreme seasonal volatility is present.
- Roti Canai is the lowest revenue item (RM 662,169) despite having the second-highest unit count (286,505 units). Its very low price (avg RM 2.43) limits revenue despite high volume.
- Laksa has both the lowest revenue (RM 655,384) and the lowest unit count (47,347), making it the least commercially successful item across both dimensions.

12. Business Recommendations

The following recommendations are grounded directly in the analytical findings above. They represent data-informed suggestions, not guaranteed outcomes.

- Leverage Mushroom Soup's high margin (82.4%) for upselling. Despite ranking 10th in revenue, it offers the best margin efficiency. Menu placement and server training to recommend it more actively could improve profitability without increasing volume.
- Investigate promotional timing. Since promotions are associated with the highest demand lift (91.8%), operators should analyse whether promotions are being run on days/items that would have been high-demand regardless. Targeted promotions on genuinely slow periods could improve their ROI.
- Plan staffing and inventory for weekends. Weekend demand is 46.2% higher than weekdays. This translates directly to staffing and ingredient requirements — understocking on weekends risks lost sales.
- Evaluate Laksa's menu position. Laksa is the lowest-revenue, lowest-volume item in the dataset. It may benefit from a price review, a menu position change, or being promoted more actively. Consider whether it complements the menu or can be replaced.
- Explore Fine Dining revenue growth through volume. Fine Dining's revenue is comparable to Food Stalls despite far fewer transactions. If capacity exists, targeted promotions or events during low-demand months (September, July) could close the gap.
- Use the demand model for inventory planning. The Random Forest model achieves $R^2 = 0.758$ on out-of-sample test data. Predicted demand scores can be used to support ingredient ordering for the next week or month, particularly for perishable items.
- Review the 502 zero-quantity rows. These represent days where items were listed but no units were sold. Understanding why (menu availability, preparation failures, data entry gaps) could reveal preventable lost-sales opportunities.

- Consider bundling low-margin, high-volume items with high-margin items. Pairing Roti Canai (high volume, lowest margin at 66.2%) with Teh Tarik (high volume, 69.8% margin) in a set meal could increase the effective margin per transaction.

13. Interactive Streamlit Dashboard

The Streamlit dashboard (restaurant_demand_analytics.py) provides a fully interactive analytics interface with four navigation tabs, global sidebar filters, and dynamically computed KPIs and insights.

13.1 Sidebar Filters

All four dashboard pages respond to the following filters applied in the sidebar:

- Date Range: start and end date selector covering the full dataset range
- Restaurant Type: multi-select (Food Stall, Cafe, Kopitiam, Casual Dining, Fine Dining)
- Menu Item: multi-select from all 14 menu items
- Meal Type: multi-select (Breakfast, Lunch, Dinner)
- Weather: multi-select (Sunny, Cloudy, Rainy)
- Show only Promoted rows: checkbox filter
- Show only Special Event rows: checkbox filter

All KPIs, charts, tables, and insights update in real time when any filter changes.

13.2 Tab 1: Executive Overview

Displays six KPI metric cards (Total Revenue, Units Sold, Gross Margin, Avg Selling Price, Restaurants, Menu Items), a monthly revenue trend line chart, a weekly demand trend chart, a Top-N items horizontal bar chart (user-adjustable slider), and a sortable item summary table.

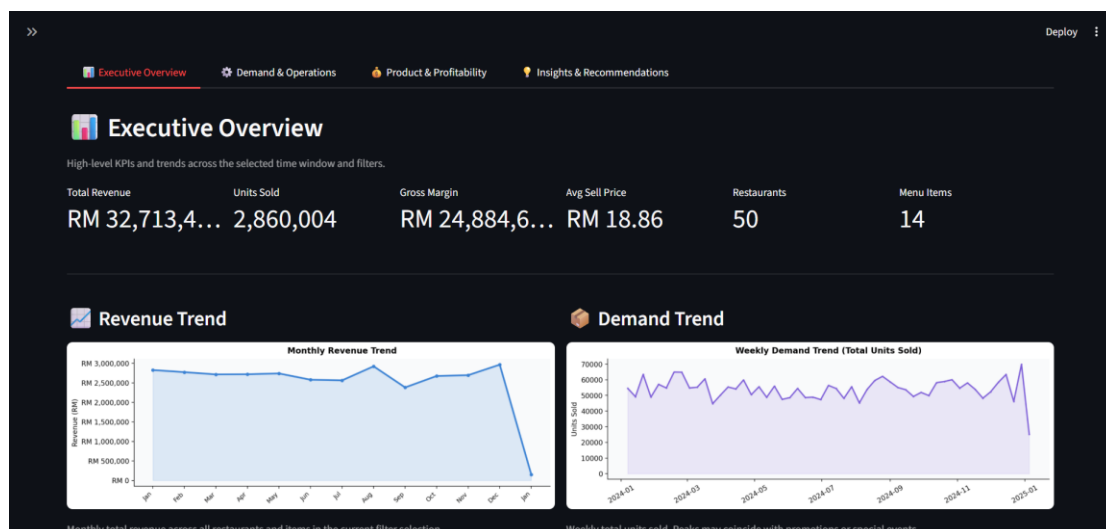


Fig 1: Executive Overview interface

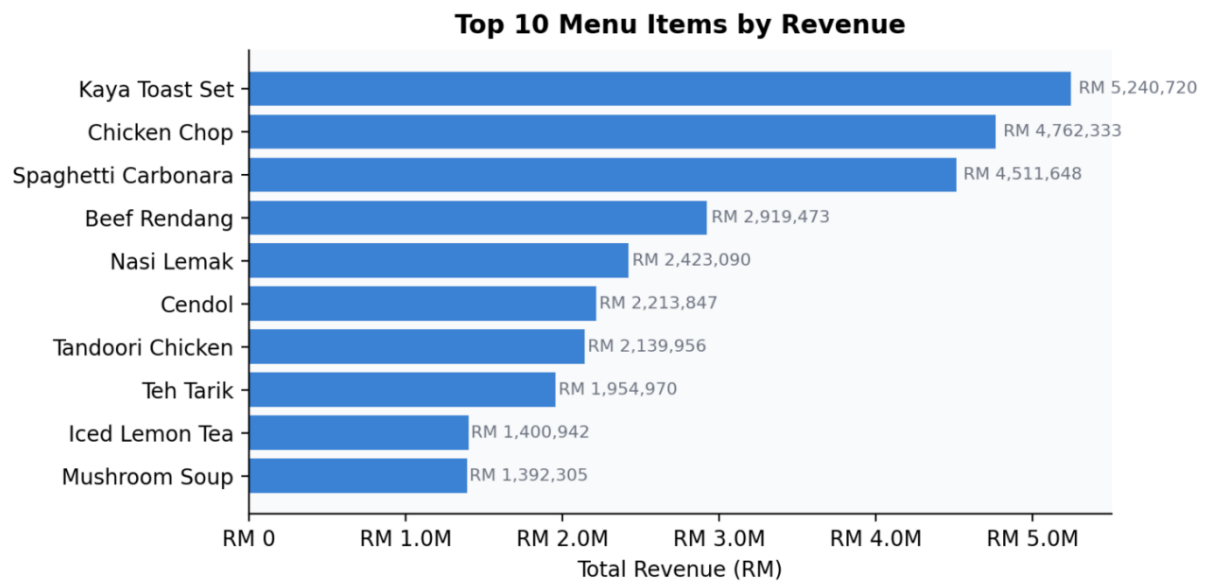


Fig 2:Top Menu Items by Revenue

Item Summary Table					
Menu Item	Total Revenue (RM)	Total Units	Avg Price (RM)	Avg Margin %	Transactions
Kaya Toast Set	5240719.63	544101	10.07	70.5	1245
Chicken Chop	4762333	109641	47.26	81.6	693
Spaghetti Carbonara	4511648.25	93114	53.48	81.5	685
Beef Rendang	2919472.66	66038	48.14	78.5	626
Nasi Lemak	2423090.12	226795	11.22	75	658
Cendol	2213847.11	304622	7.86	72.6	1219
Tandoori Chicken	2139956.04	91048	25.53	71.2	726
Teh Tarik	1954969.69	643076	3.16	69.8	1147
Iced Lemon Tea	1400942.3	235325	6.28	79.6	677
Mushroom Soup	1392304.64	70164	21.46	82.4	671

Fig 3: Item Summary Table

13.3 Tab 2: Demand & Operations

Displays the day-of-week average demand chart, meal-type demand chart, promotion effect bar chart, special event effect bar chart, weather effect bar chart, and the machine learning Actual vs Predicted demand scatter plot with MAE/RMSE/R2 metric cards. Each chart includes a caption explaining what the chart shows and any caveats.

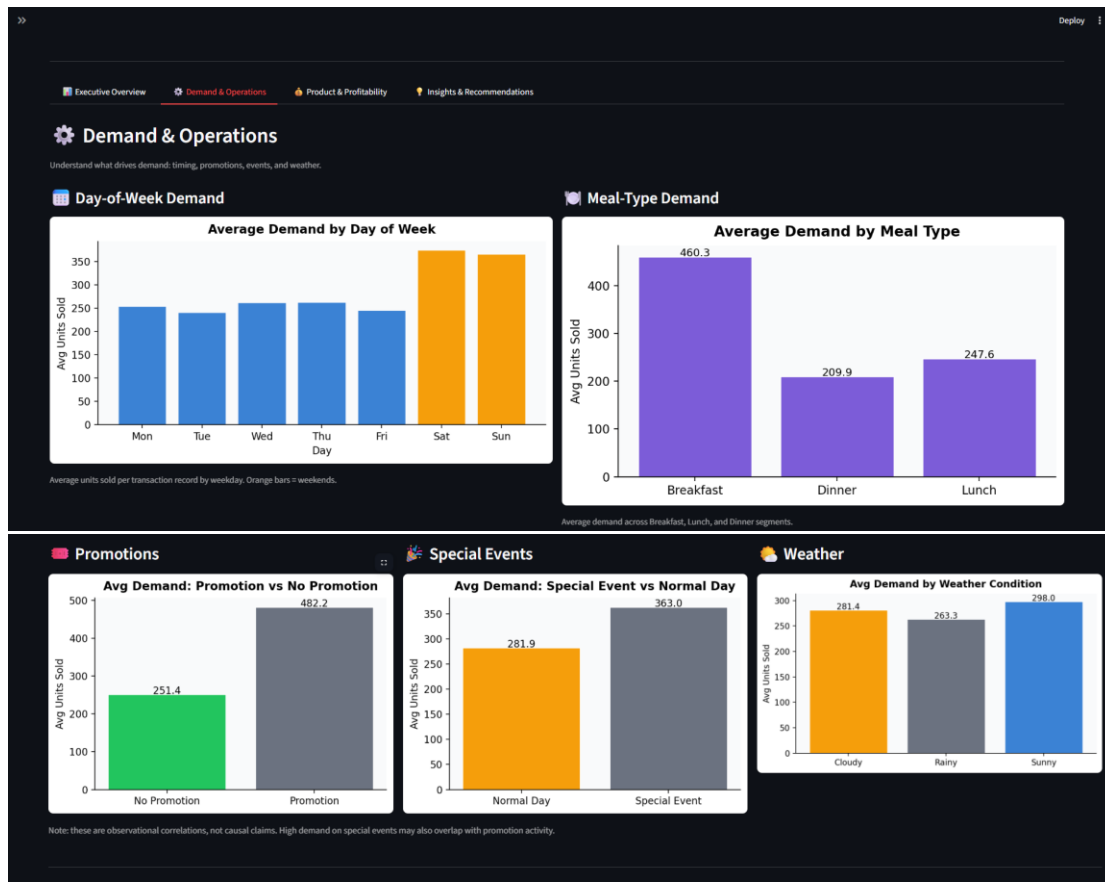


Fig 4: Graphs

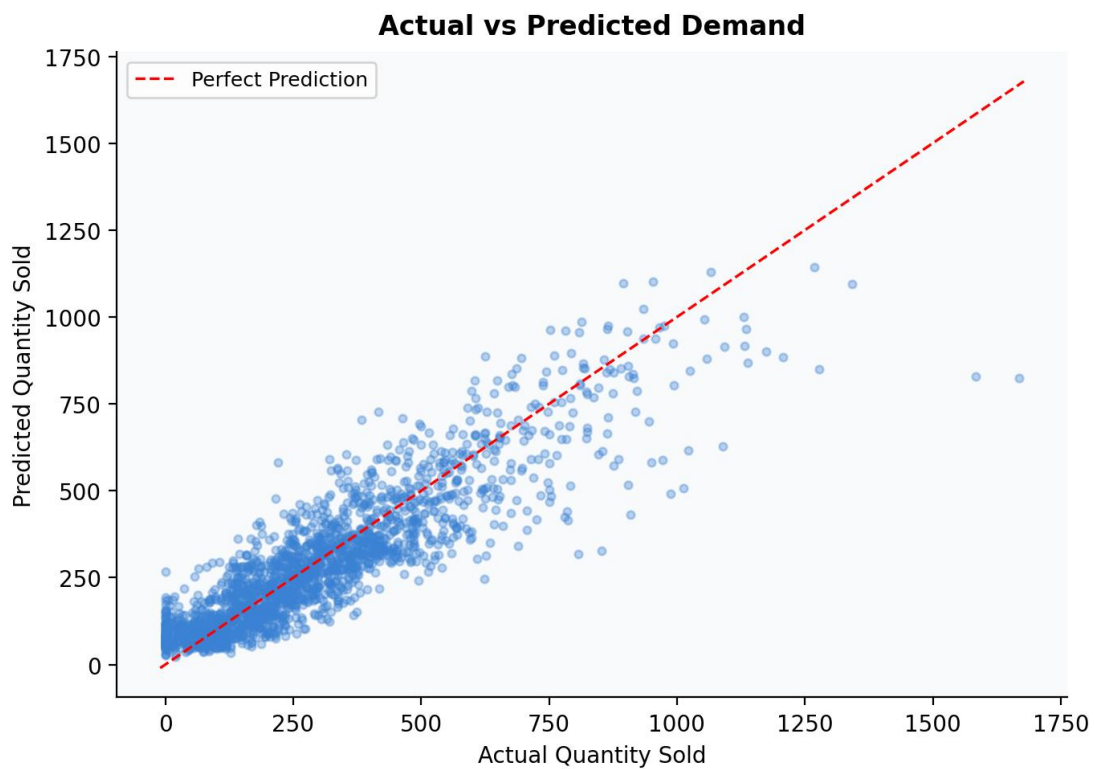


Fig 5: Machine Learning — Demand Prediction

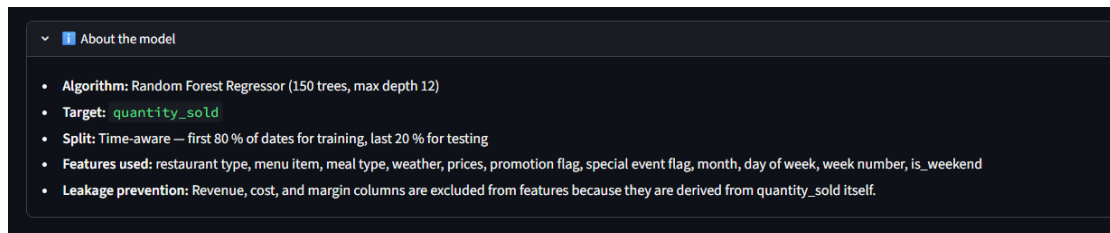


Fig 6: Model Description

13.4 Tab 3: Product & Profitability

Displays revenue by restaurant type bar chart, average gross margin % by menu item horizontal bar chart (with colour-coded thresholds: green $\geq 60\%$, yellow 40-60%, red $< 40\%$), price vs quantity scatter chart coloured by gross margin, market price vs actual selling price scatter chart, a low-performing items table (bottom 25th percentile by revenue), and a full profitability table in an expandable section.

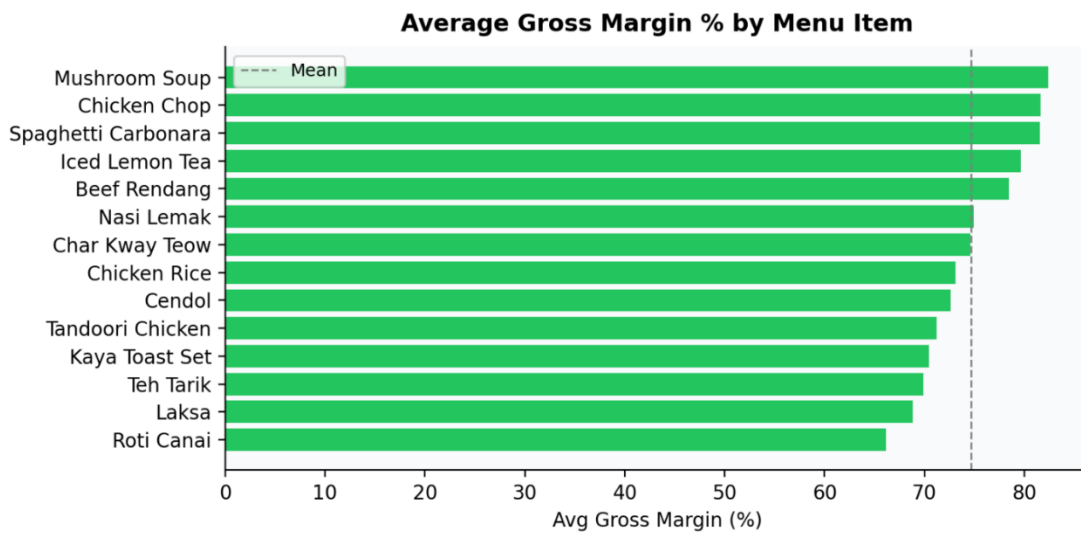


Fig 7: Revenue by Restaurant Type

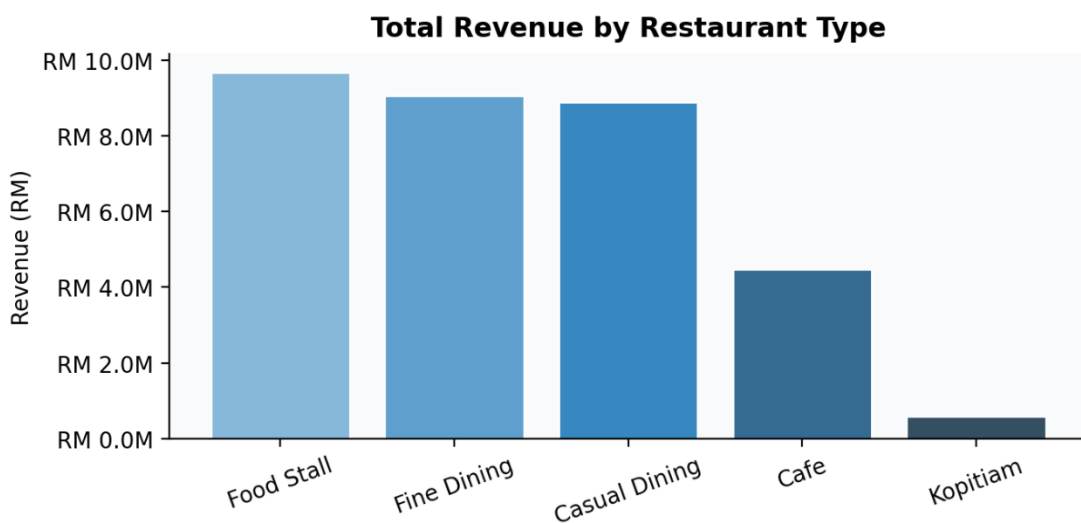


Fig 8: Gross Margin % by Menu Item

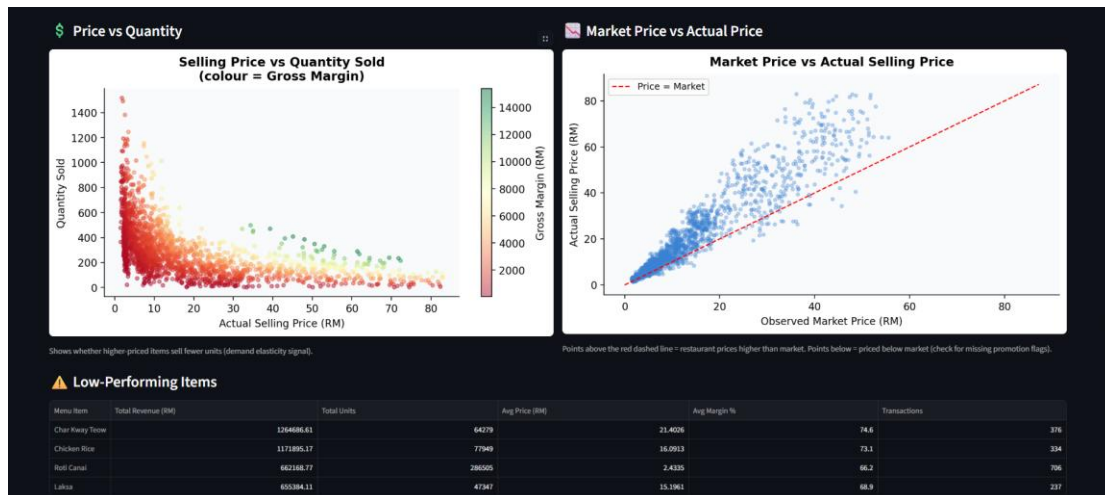


Fig 9: Product & Profitability

13.5 Tab 4: Insights & Recommendations

Dynamically generates 6-7 business insights from the filtered data at runtime (no hard-coded text). Includes automatic recommendation bullets for the top revenue item, best margin item, lowest revenue item, promotion effectiveness, and zero-quantity row count — all computed from the actual data matching the current filter selection. This section updates when any filter changes.

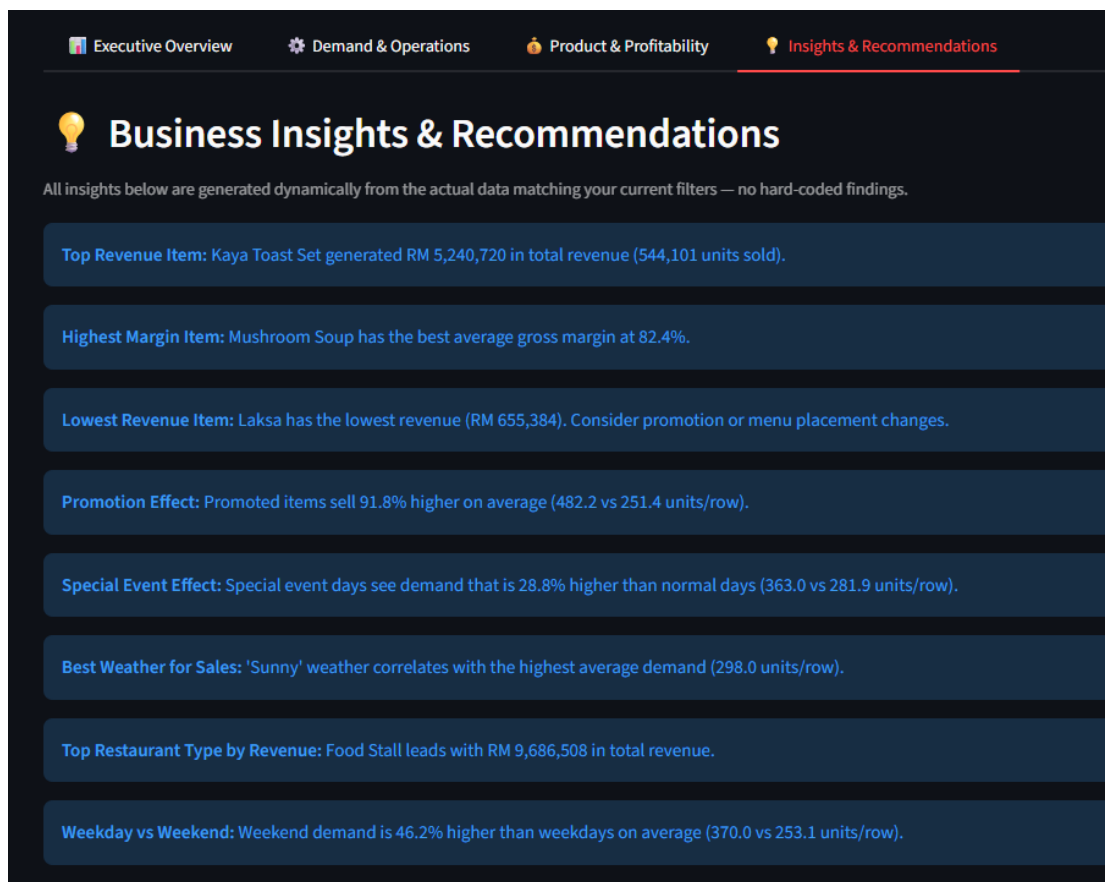


Fig 10: Insights

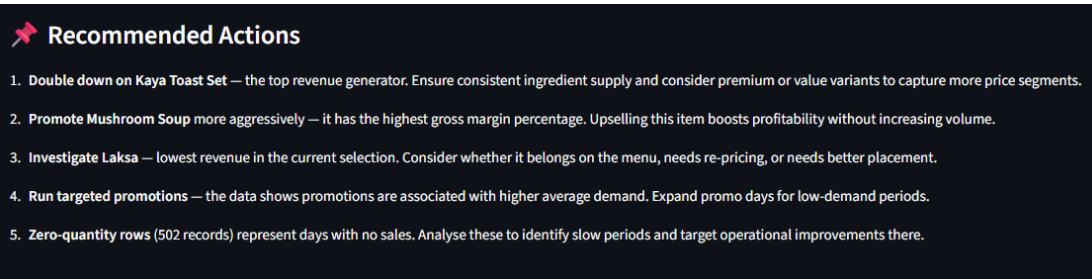


Fig 11: Recommendations

14. Limitations

- **Observational data only:** All reported associations (promotions, weather, special events) are correlations observed in historical transaction records. This project does not establish causal relationships. Confounding variables (e.g., promotions more likely to be applied on high-traffic days) may inflate observed effect sizes.
- **Ingredient cost margin only:** The `estimated_gross_margin` and `margin_percentage` features are calculated from `typical_ingredient_cost` only. They do not account for labour costs, rent, utilities, packaging, or any other operating expenses. Actual net margins will be lower.
- **Duplicate transaction keys:** 366 groups of rows (755 total) share the same date + restaurant + item + meal combination. These have been retained but not resolved. If they represent duplicate data entry rather than multiple sittings, aggregated totals may be slightly inflated.
- **Zero-quantity rows (502):** The reason why 5% of rows have `quantity_sold = 0` is unknown from the data alone. They could represent menu items available but unsold, data entry placeholders, or other operational events.
- **No external validation:** The dataset represents a single source. There is no independent validation against external benchmarks or ground-truth restaurant performance data.
- **Feature importance caveats:** The Random Forest model assigns the highest importance to `actual_selling_price` (0.5662). This largely reflects the structural relationship between price and item type (expensive items sell fewer units), not a simple price elasticity effect.
- **Single year of data:** The dataset covers exactly one calendar year. Longer-term trends, year-over-year growth, or multi-year seasonality cannot be assessed.

15. Technologies Used

Technology	Version Used	Purpose
Python	3.11	Core programming language
Pandas	1.5.3	Data loading, cleaning, feature engineering, EDA aggregations
NumPy	1.24.3	Numerical operations, safe division, array handling
Matplotlib / Seaborn	3.7+ / 0.12+	All 14 static visualisation charts

Technology	Version Used	Purpose
Scikit-learn	1.3+	RandomForestRegressor, LabelEncoder, MAE/RMSE/R2 metrics
Streamlit	1.58.0	Interactive dashboard: tabs, filters, KPI cards, charts
Python standard library	3.11	sys, os, warnings — no non-standard system dependencies

16. Conclusion

This project demonstrates a complete, production-ready data analytics pipeline applied to a real restaurant sales dataset. Starting from raw CSV data, the project delivers:

- A clean, well-validated dataset with documented quality issues and applied cleaning decisions
- A rich set of engineered features covering revenue, margin, pricing, and temporal dimensions
- Accurate KPIs (RM 32.7M total revenue, 2.86M units sold, 74.2% average gross margin)
- Comprehensive EDA identifying the key drivers of demand: promotions, weekends, meal type, special events, and weather
- A Random Forest demand prediction model achieving $R^2 = 0.758$, MAE = 78.53 units on held-out future data
- Dynamic, filter-driven business insights generated directly from the data at runtime
- A four-tab interactive Streamlit dashboard accessible to non-technical stakeholders

The project meets all IBM SkillsBuild Final Project requirements: data loading, cleaning, feature engineering, KPI analysis, EDA, machine learning, model evaluation, demand prediction, business insights, and an interactive dashboard — all implemented in a single Python file.

Link of the dataset: <https://www.kaggle.com/datasets/alexandchen/restaurant-sales-report-2024-2025>